Whitmore Capital Partners SAA/TAA Mandate

FIN 496 Foundation Project

- External consultant recommendation for a fictional $1.8B single-family office.
- Pipeline is fully reproducible from `python taa_project/main.py`.
- Final run mode: --no-timesfm.

Mandate & IPS Highlights

- Target 8% return with 15% volatility ceiling and 25% max drawdown tolerance.
- Core must stay at or above 40%, Satellite at or below 45%, Non-Traditional at or below 20%.
- No shorts, no hard-coded risk-off safe haven, and 5 bps round-trip cost applied at each rebalance.

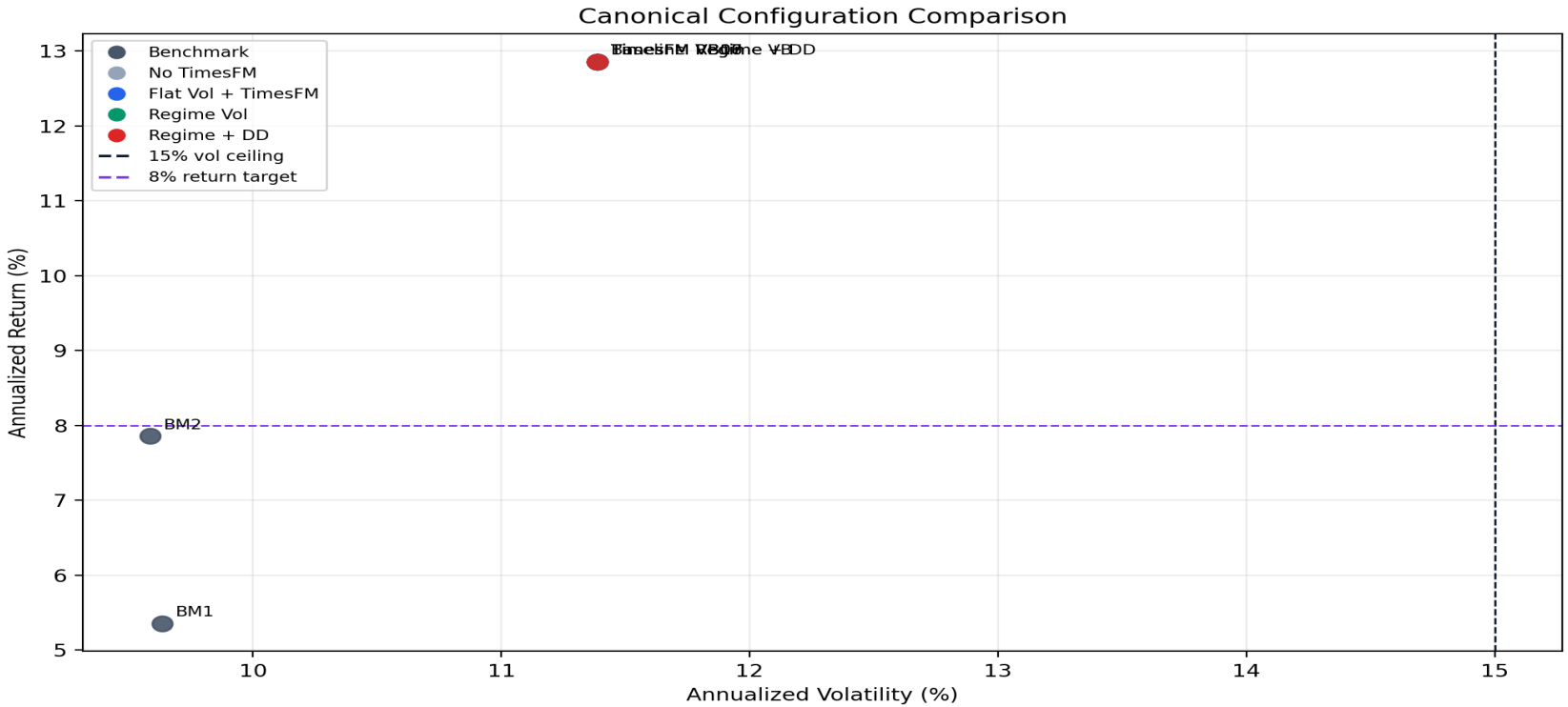
SAA Method Choice

- Chosen SAA method: constrained risk parity.
- Reason: it clears the 8% return mandate while staying below the 15% volatility ceiling without leaning on unstable mean forecasts.
- Alternatives were evaluated side-by-side in the diagnostics notebook and `saa_method_comparison.csv`.

TAA Signal Architecture

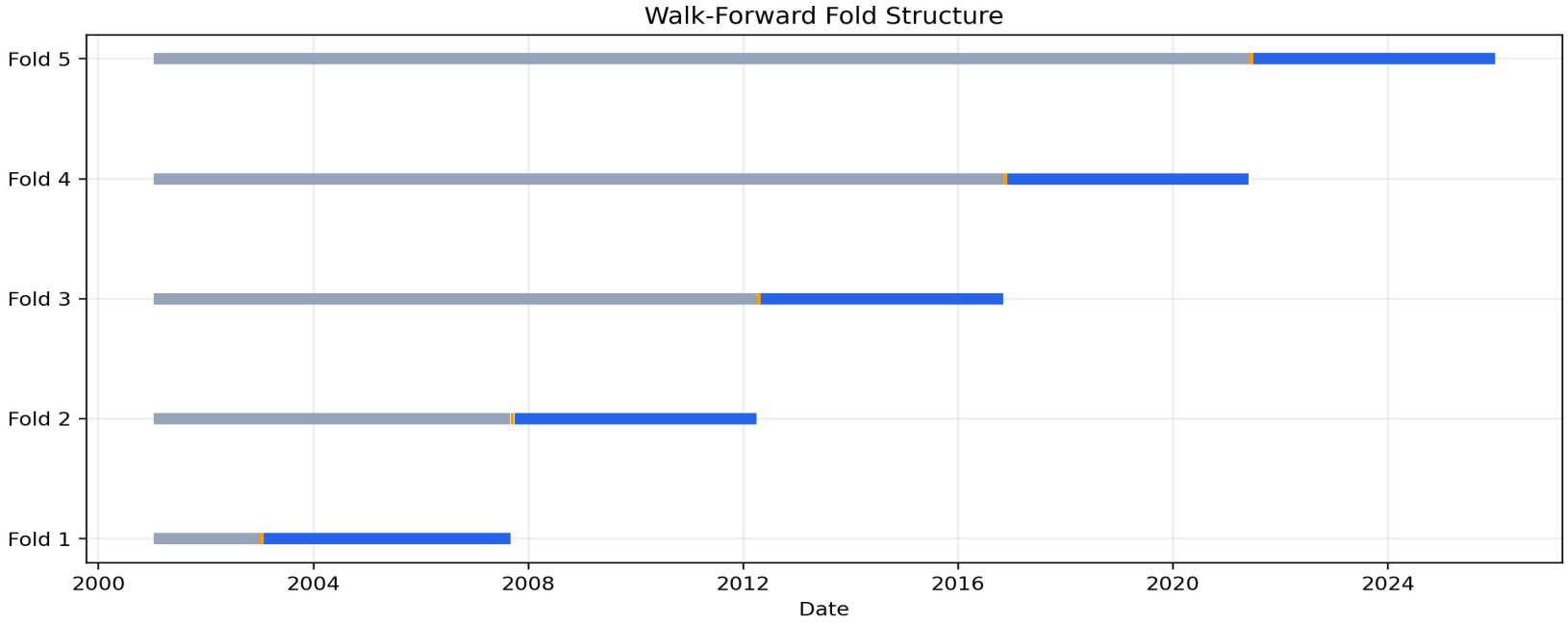
- Lagged macro -> 3-state HMM regime probabilities.
- Daily price histories -> Faber trend + ADM momentum.
- Optional TimesFM forecast -> direction and volatility input.
- Signal ensemble -> cvxpy optimizer inside TAA bands with turnover and volatility controls.

Risk Overlays & Vol-Budget Sweep



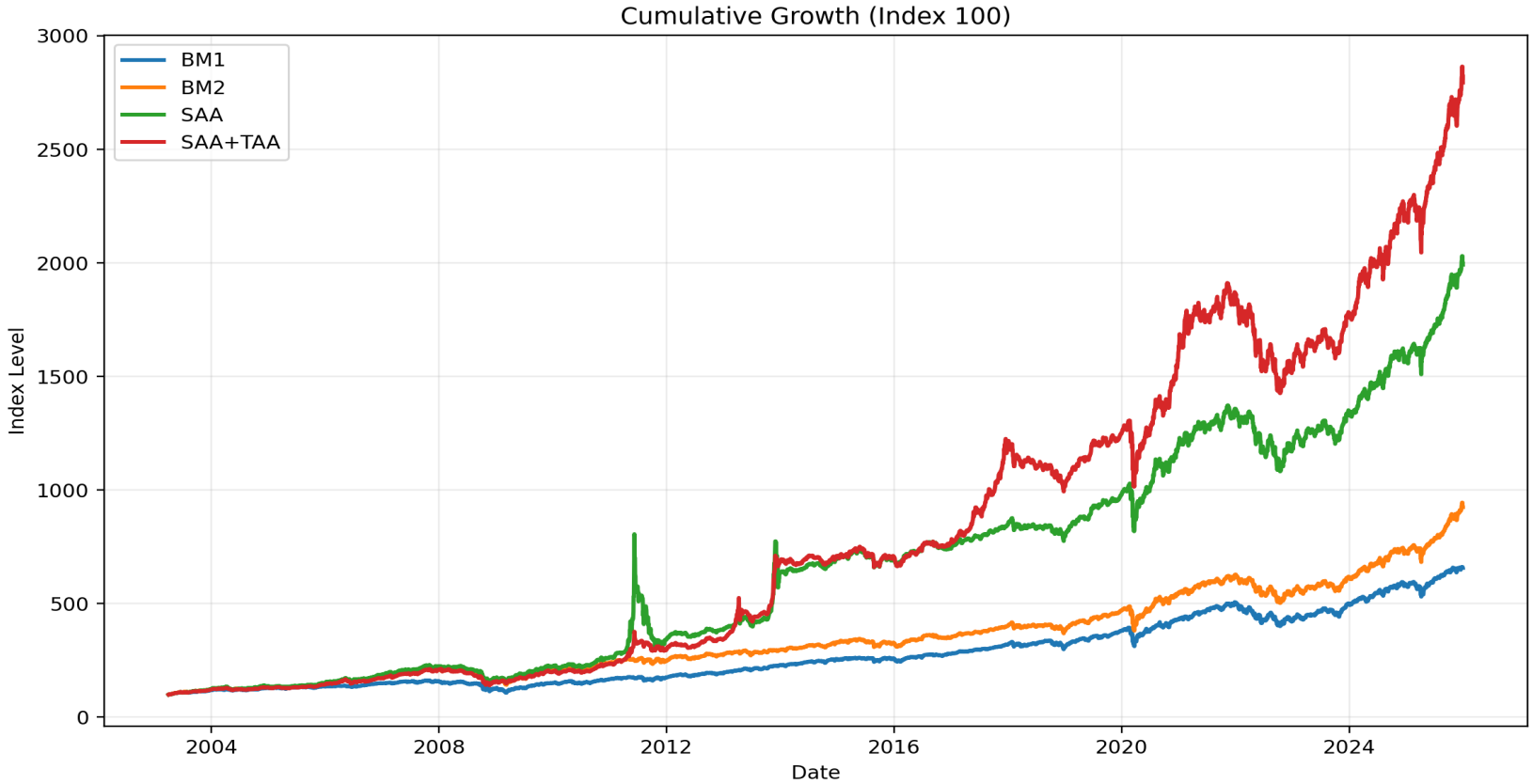
- Selected configuration: baseline (Baseline).
- Best tested max drawdown: -34.55% across 6 canonical configurations.
- All six canonical variants converged to the same realized OOS portfolio path, so the baseline was retained as the simplest tied winner.
- Overlays only tighten the risk envelope; they do not hard-code a safe-haven allocation.

Walk-Forward Design



Deflated Sharpe Ratio: 1.000 across 4 disclosed TAA trials.

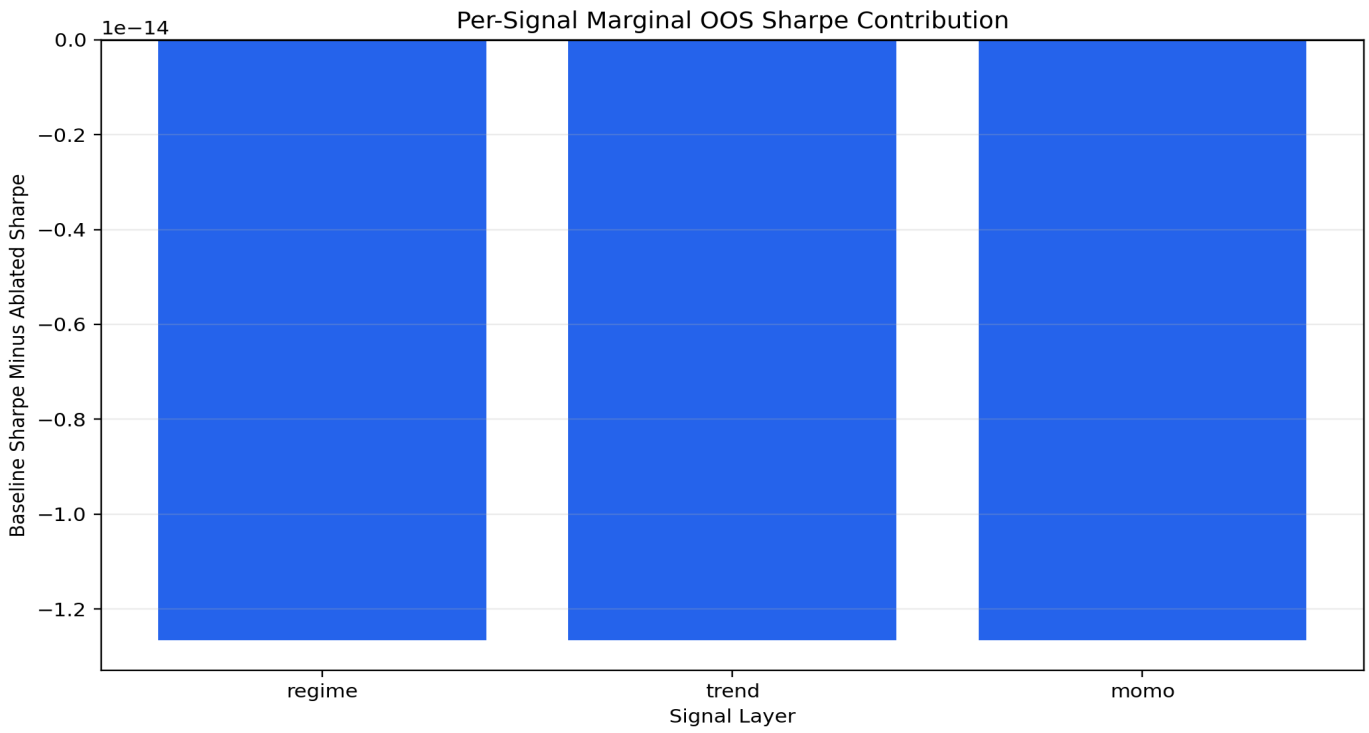
Cumulative Growth



Metrics Table

portfolio	annualized_return	annualized_volatility	Sharpe	Sortino	Calmar	max_drawdown
BM1	0.054	0.096	0.349	0.489	0.158	-0.339
BM2	0.079	0.096	0.611	0.854	0.223	-0.352
SAA	0.088	0.144	0.474	0.715	0.143	-0.618
SAA+TAA	0.129	0.114	0.954	1.391	0.372	-0.345

Attribution



- Separate decomposition delivered for SAA vs BM2 and TAA vs SAA/BM1/BM2.
- Signal value is measured with leave-one-out OOS reruns, not static coefficient inspection.

Limitations

- HMM states are statistical and can drift.
- TimesFM is optional and not finance-native.
- Covariance shrinkage and optimizer penalties remain engineering choices, not immutable truths.

Recommendation

- SAA+TAA annualized return: 12.86% versus 8.83% for SAA and 7.86% for BM2.
- Submitted configuration: baseline. Max drawdown misses the -25% IPS tolerance at -34.55%.
- Track turnover and stressed-regime attribution before expanding risk budget to the overlay.

Q&A;

- Key files: `taa_project/main.py`, `taa_project/analysis/reporting.py`, `taa_project/report/build_report.py`.
- Key artifacts: report PDF, deck PDF, diagnostics notebook, trial ledger, IPS audit.